

CSC1109 – Data (Analytics) at Speed and Scale

Alessandra Mileo

alessandra.mileo@dcu.ie

Finding Friends

- “You and Joe have 14 friends in common”
- Does not change that often
- Can precompute (e.g. once a day) with MapReduce

Input (Key, Value): (User, [List of friends])

MAP

REDUCE

Output (Key, Value): (User1 User2, [List of common friends])

Finding Friends: MAPPER

Input: **(User, [List of friends])**

```
1 (A, [B C D])
2 (B, [A C D E])
3 (C, [A B D E])
4 (D, [A B C E])
5 (E, [B C D])
```

For 1 (A, [B C D])
the Mapper generates:

```
(A B, [B C D])
(A C, [B C D])
(A D, [B C D])
```

Key (for each friend)

Friend Person

Sorted in order

Appearing once as
(A B = B A)

Value (for each key)

[list of friends]

Finding Friends: MAPPER

Input: **(User, [List of friends])**

```
1 (A, [B C D])
2 (B, [A C D E])
3 (C, [A B D E])
4 (D, [A B C E])
5 (E, [B C D])
```

For 2 (B, [A C D E])
the Mapper generates:

```
(A B, [A C D E])
(B C, [A C D E])
(B D, [A C D E])
(B E, [A C D E])
```

Key (for each friend)

Friend Person

Sorted in order

Appearing once as
(A B = B A)

Value (for each key)

[list of friends]

Finding Friends: MAPPER

Input: **(User, [List of friends])**

```
1 (A, [B C D])
2 (B, [A C D E])
3 (C, [A B D E])
4 (D, [A B C E])
5 (E, [B C D])
```

For 3 (C, [A B D E])
the Mapper generates:

```
(A C, [A B D E])
(B C, [A B D E])
(C D, [A B D E])
(C E, [A B D E])
```

Key (for each friend)

Friend Person

Sorted in order

Appearing once as
(A B = B A)

Value (for each key)

[list of friends]

Finding Friends: MAPPER

Input: **(User, [List of friends])**

```
1 (A, [B C D])
2 (B, [A C D E])
3 (C, [A B D E])
4 (D, [A B C E])
5 (E, [B C D])
```

For 4 (D, [A B C E])
the Mapper generates:

```
(A D, [A B C E])
(B D, [A B C E])
(C D, [A B C E])
(D E, [A B C E])
```

Key (for each friend)

Friend Person

Sorted in order

Appearing once as
(A B = B A)

Value (for each key)

[list of friends]

Finding Friends: MAPPER

Input: **(User, [List of friends])**

```
1 (A, [B C D])
2 (B, [A C D E])
3 (C, [A B D E])
4 (D, [A B C E])
5 (E, [B C D])
```

For 5 (E, [B C D])
the Mapper generates:

```
(B E, [B C D])
(C E, [B C D])
(D E, [B C D])
```

Key (for each friend)

Friend Person

Sorted in order

Appearing once as
(A B = B A)

Value (for each key)

[list of friends]

Finding Friends: Group-by-Key

Key

Value

Friend Person

[list of friends]

(Friend Person, [List of friends 1]...[List of friends N])

(A B, [A C D E] [B C D])

(A C, [A B D E] [B C D])

(A D, [A B C E] [B C D])

(B C, [A B D E] [A C D E])

(B D, [A B C E] [A C D E])

(B E, [A C D E] [B C D])

(C D, [A B C E] [A B D E])

(C E, [A B D E] [B C D])

(D E, [A B C E] [B C D])

Finding Friends: REDUCE

Key		Value
Friend Person		[list of friends]
(Friend Person, [List of friends 1]...[List of friends N])		
1	(A B, [A C D E] ∩ [B C D])	(A B, [C D])
2	(A C, [A B D E] ∩ [B C D])	(A C, [...])
3	(A D, [A B C E] ∩ [B C D])	(A D, [...])
4	(B C, [A B D E] ∩ [A C D E])	(B C, [...])
5	(B D, [A B C E] ∩ [A C D E])	(B D, [...])
6	(B E, [A C D E] ∩ [B C D])	(B E, [...])
7	(C D, [A B C E] ∩ [A B D E])	(C D, [...])
8	(C E, [A B D E] ∩ [B C D])	(C E, [...])
9	(D E, [A B C E] ∩ [B C D])	(D E, [...])
(Friend Person, ∩([List of friends 1]...[List of friends N]))		

Finding Friends: REDUCE

Key		Value
Friend Person		[list of friends]
(Friend Person, [List of friends 1]...[List of friends N])		
1	(A B, [A C D E] ∩ [B C D])	(A B, [C D])
2	(A C, [A B D E] ∩ [B C D])	(A C, [B D])
3	(A D, [A B C E] ∩ [B C D])	(A D, [B C])
4	(B C, [A B D E] ∩ [A C D E])	(B C, [A D E])
5	(B D, [A B C E] ∩ [A C D E])	(B D, [A C E])
6	(B E, [A C D E] ∩ [B C D])	(B E, [C D])
7	(C D, [A B C E] ∩ [A B D E])	(C D, [A B E])
8	(C E, [A B D E] ∩ [B C D])	(C E, [B D])
9	(D E, [A B C E] ∩ [B C D])	(D E, [B C])
(Friend Person, ∩([List of friends 1]...[List of friends N]))		

LAB: Where are we?

Running MapReduce via MRJob in Python:

- <https://csc1109.readthedocs.io/en/latest/lab1/>

Hive

- <https://github.com/CA4022/Lab1-HadoopMR>

Finding Friends and Beyond

- Findinig Friends in JAVA:

<https://github.com/vamseekrishnakattika/Mutual-Friends-using-Hadoop-Map-Reduce>

- More examples:

<https://highlyscalable.wordpress.com/2012/02/01/mapreduce-patterns/>

- MR native (Java/Pythin): can we make it easier?
 - Hive
 - Pig

The Hadoop Ecosystem recap

- Data Factory vs Data Warehouse
- Where does *HIVE* stand?
- Where does *PIG* stand?
- ELT or ETL?